

Matthew Pidden

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Education

University of Bristol

Graduating July 2025

Bachelor of Science in Computer Science (with First Class Honours, 81%)

Bristol, UK

- **Third Year Classes:** Final Dissertation Project: 91, Machine Learning: 87, Computer Vision: 83, Artificial Intelligence: 78, High Performance Computing: 71
- **Second Year Classes:** Security and Operating Systems: 95, Programming Languages and Computation: 87, Software Engineering Project: 85, Concurrent and Distributed Programming: 80, Algorithms II: 74

Experience

Global Expeditions Group

October 2023 – Present

Software Engineer (Internship transitioned into Part Time)

Remote

- Built and maintained two cross-platform Flutter applications for student blogs and staff portal access, both supporting offline use in remote environments.
- Integrated with WordPress and internal APIs via REST, enabling seamless content access and participant data retrieval.
- Implemented features such as offline sync, image editing/compression, and custom UI for blog and data workflows.
- Deployed to iOS, macOS, and Android with regular updates and operational support.

University of Bristol

September 2024 – May 2025

Lead Teaching Assistant (Part Time)

Bristol, UK

- Led weekly formative feedback sessions and provided technical mentorship for student teams in the 2nd Year Software Engineering Project module, supporting group development and delivery.
- Delivered lab support in the Concurrent and Distributed Programming module, assisting students with Go concurrency concepts, debugging, and practical exercises.

Projects

The Revenger | C++, Python, Arduino Due, Custom PCB, OpenCV, Fusion 360, 3D Printing, Latex

- Designed and built the world's fastest 4x4 Rubik's Cube solving robot, achieving a World Record time of **45.305 seconds** (previous: 1:18.68).
- Engineered a fully custom mechanical system with 10 stepper motors, unique claw and corner grabbers, and precision control via a self-designed PCB and Arduino firmware.
- Implemented a vision pipeline using Python and OpenCV with dual webcams to scan cube faces and trigger a real-time solving algorithm.
- Developed multiple solving methods, including a custom implementation of an 8-step algorithm with pattern databases, IDA* search, and masked BFS optimisations.

The Picks | Flutter, Firebase Hosting, Firestore, Google Cloud Functions, JavaScript

- Built a cross-platform Premier League prediction app and website with a shared Firebase backend, used by 50+ users globally.
- Modernized a 20-year-old family game with real-time scoring, global and private leaderboards, and automated ranking via Premier League API integration.
- Delivered user-specific push notifications for gameweek reminders and results, enhancing engagement and retention.

Santa Tracker | Flutter, Firebase, RevenueCat, Google AdMob, REST APIs

- I developed and marketed a viral Christmas app from scratch, reaching 35,235 downloads and ranking #8 overall and #5 in Entertainment on the UK App Store.
- Introduced rewarded ads for premium feature trials, significantly boosting in-app purchase conversions (598 purchases, \$3,245 total revenue).

Technical Skills

Languages: C++, Python, C, Dart, JavaScript, Go, Java, Swift, YAML

Technologies: Git, GitHub, Firebase (Hosting, Firestore, Auth), Flutter, React, Android SDK, iOS SDK, REST APIs, RevenueCat, OpenCV, AdMob, AWS, Microcontrollers, Arduino, PCB Design, 3D Printing